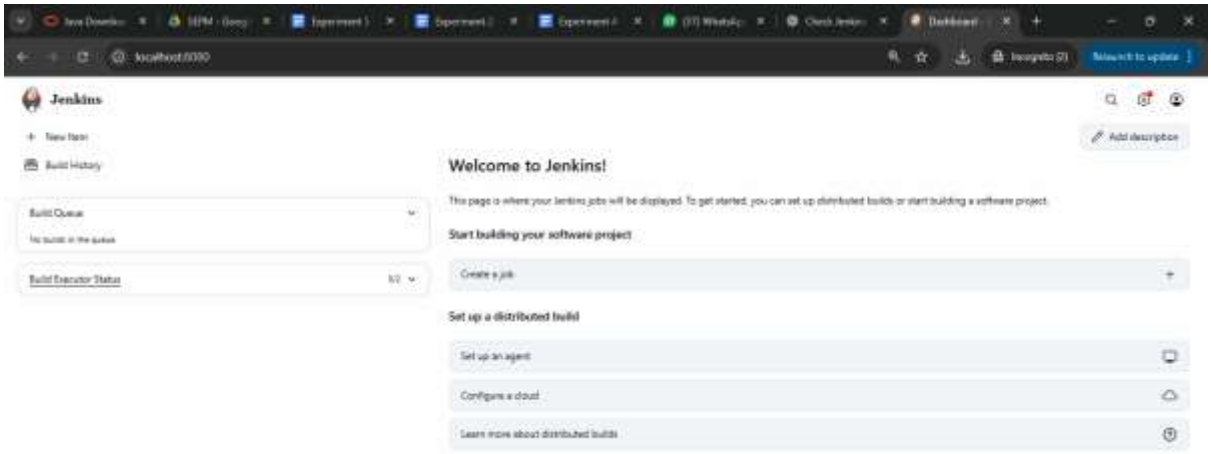
Experiment 5

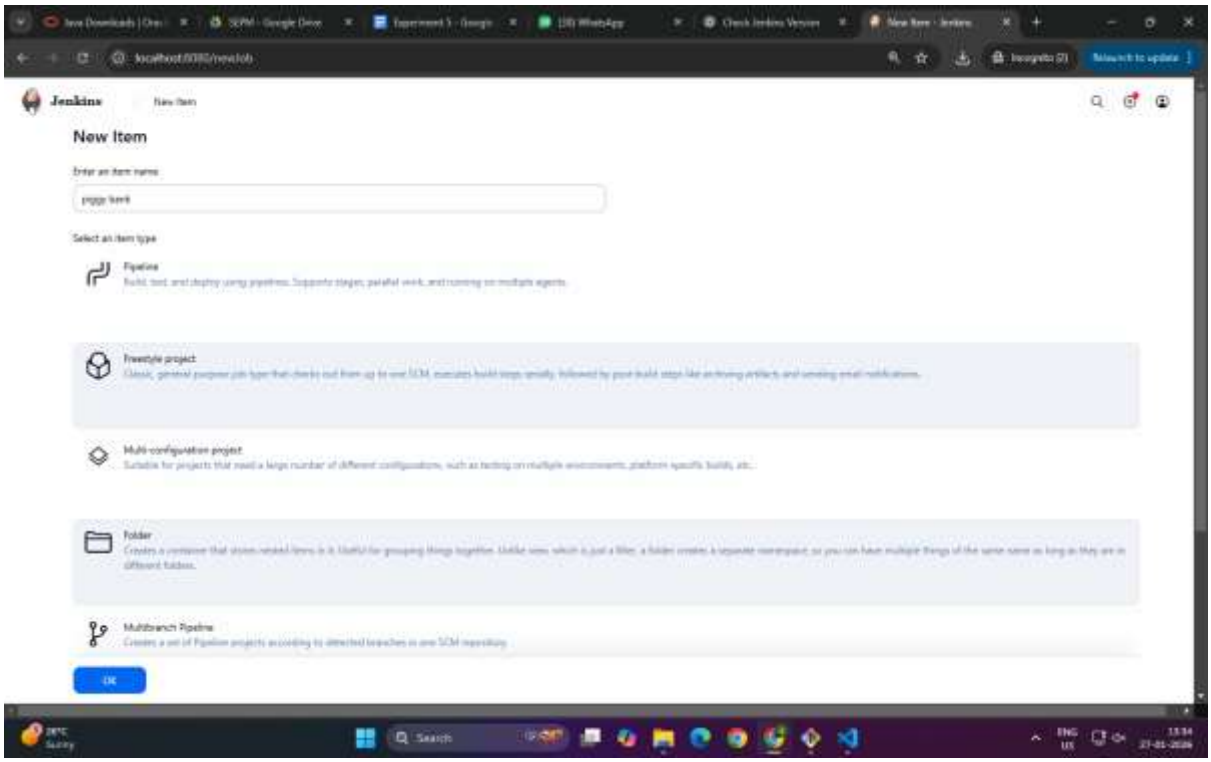
Aim - To Build the pipeline of jobs using Maven / Gradle / Ant in Jenkins, create a pipeline script to Test and deploy an application over the tomcat server

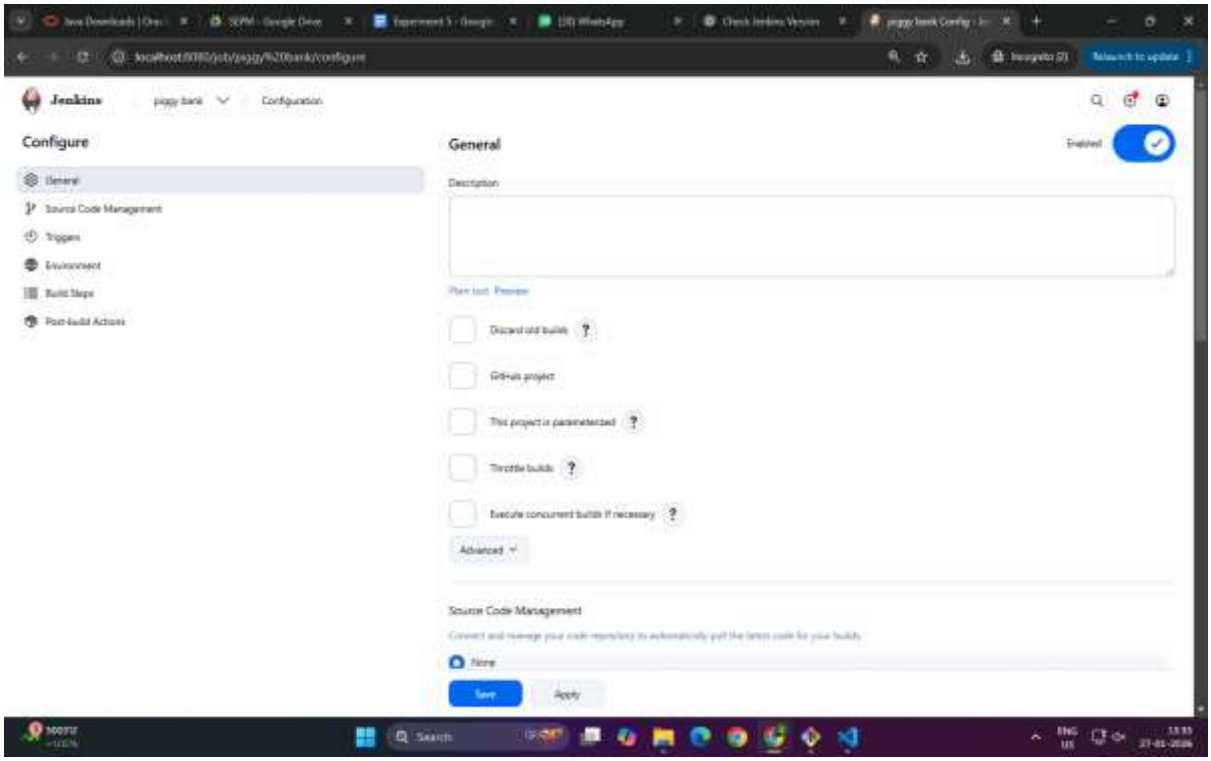
Programming in Jenkins:
Continuous Integration is a software development practice where members of a team integrate their
work frequently, usually each person integrates at least daily leading to multiple integrations per day.
Each integration is verified by an automated build (including test) to detect integration errors as
quickly as possible.” In simple way, Continuous integration (CI) is the practice of frequently building
and testing each change done to your code automatically.
Jenkins is a self-contained, open-source automation server which can be used to automate all sorts of
tasks related to building, testing, and delivering or deploying software.
Our first job will execute the shell commands. The freestyle project provides enough options and
features to build the complex jobs that you will need in your projects.
Example 1
Example 1.1: Deploying a freestyle app in Jenkins
Creating a job:
Naming the job and setting it as freestyle:

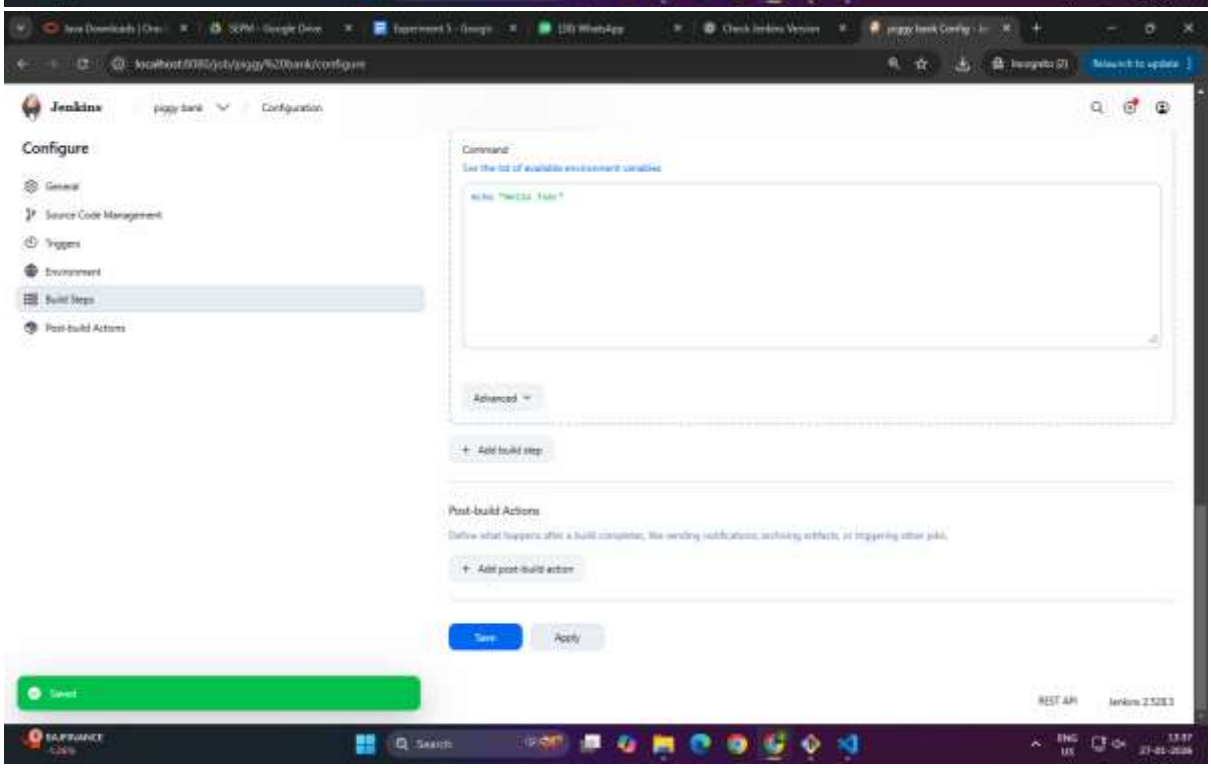
Selecting build type as “Execute shell”:
Entering a simple command for the shell execution:
Applying and saving the project configuration:
Building the project:
Console output (after building):
Example 1.2: Taking parameters through files
Contents of script example1.cmd:
Executing script example1.cmd on the terminal:
Modifying the Jenkins project to execute the script while supplying required parameters:
Console output after building the modified project:
Example 2
Example 2.1: Running a Java program under Jenkins
Creating a simple Java program:
Compiling and running the program on the terminal:
Creating a new freestyle project:
Configure new project:
Console output after building:

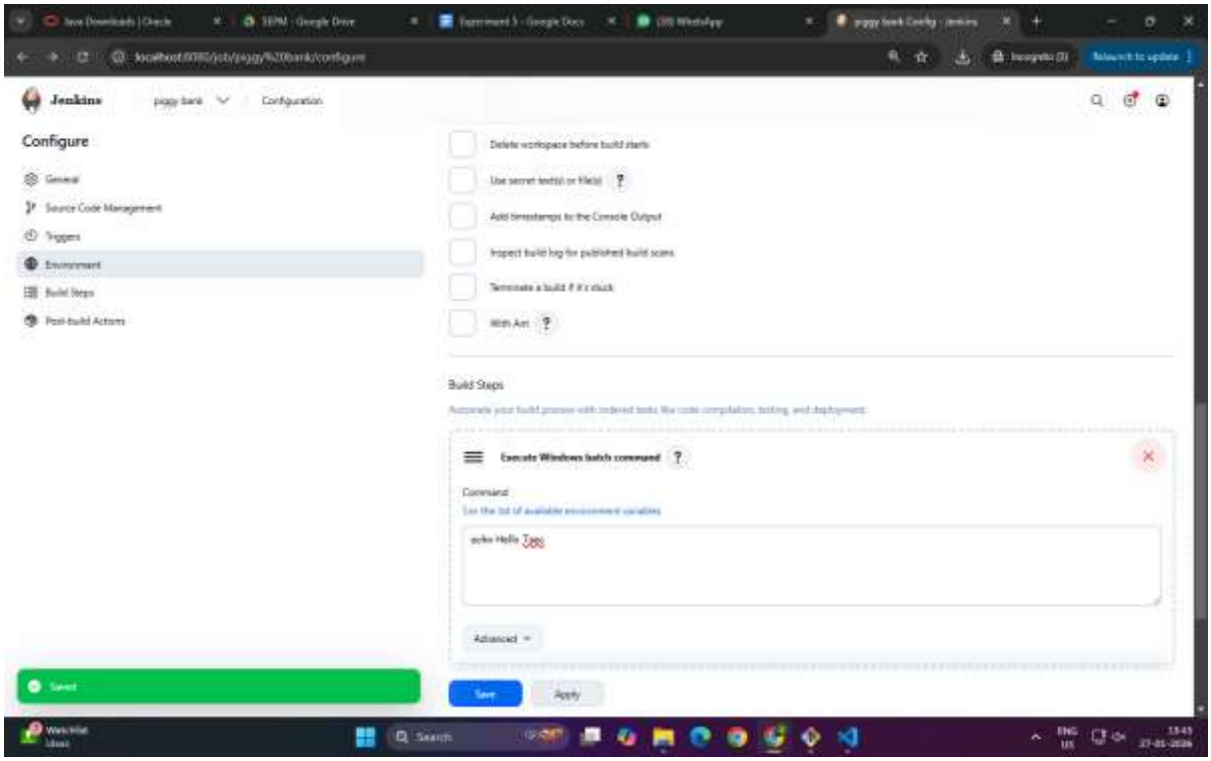
Example 3
Example 3.1: Parameterise build
Creating a new freestyle project:
Enabling parameterisation and adding a String parameter:
Configuring the string parameter as Fname:
Adding a choice parameter and configuring it as City with the following choices:
Creating a script which takes 2 arguments for name and city:
Configuring build steps:
Entering parameters for build:
Console output after building:
Example 3.2: Running a Java program with parameters
Creating a Java program with an input argument:
Testing the program on the terminal:
Creating a new freestyle project:
Parameterise the project by adding a string parameter as follows:
Configure the build steps:
Entering the parameter for the build:
Console output after building:
Example 5
Example 5.1: Running a Python program
Creating a simple Python script:
Running the Python script on the terminal:
Creating a new freestyle project:
Parameterising the project with a string parameter as follows:
Configuring the build steps:
Setting the parameter for the build:
Console output after building:
Conclusion: Thus, we have successfully studied Continuous Integration and installed, configured, and understood programming with Jenkins.

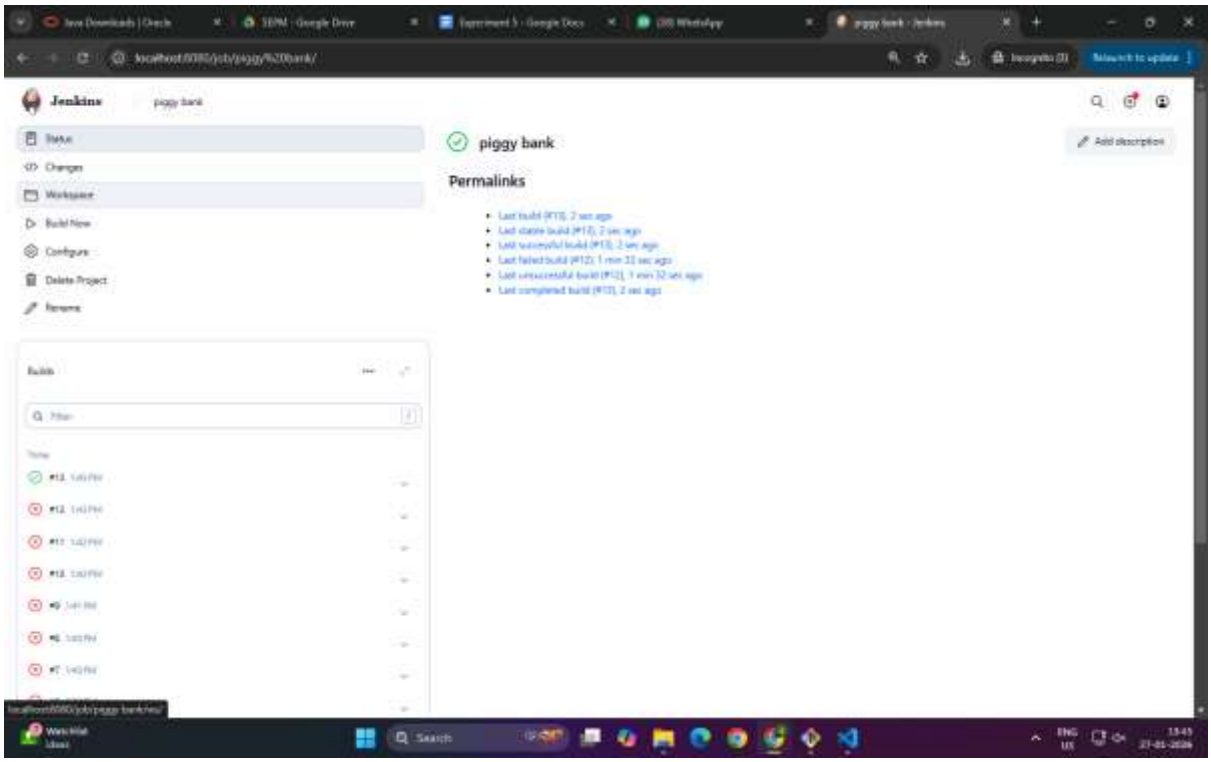

Jenkins
Welcome to Jenkins!
Start building your software project
Set up a distributed build

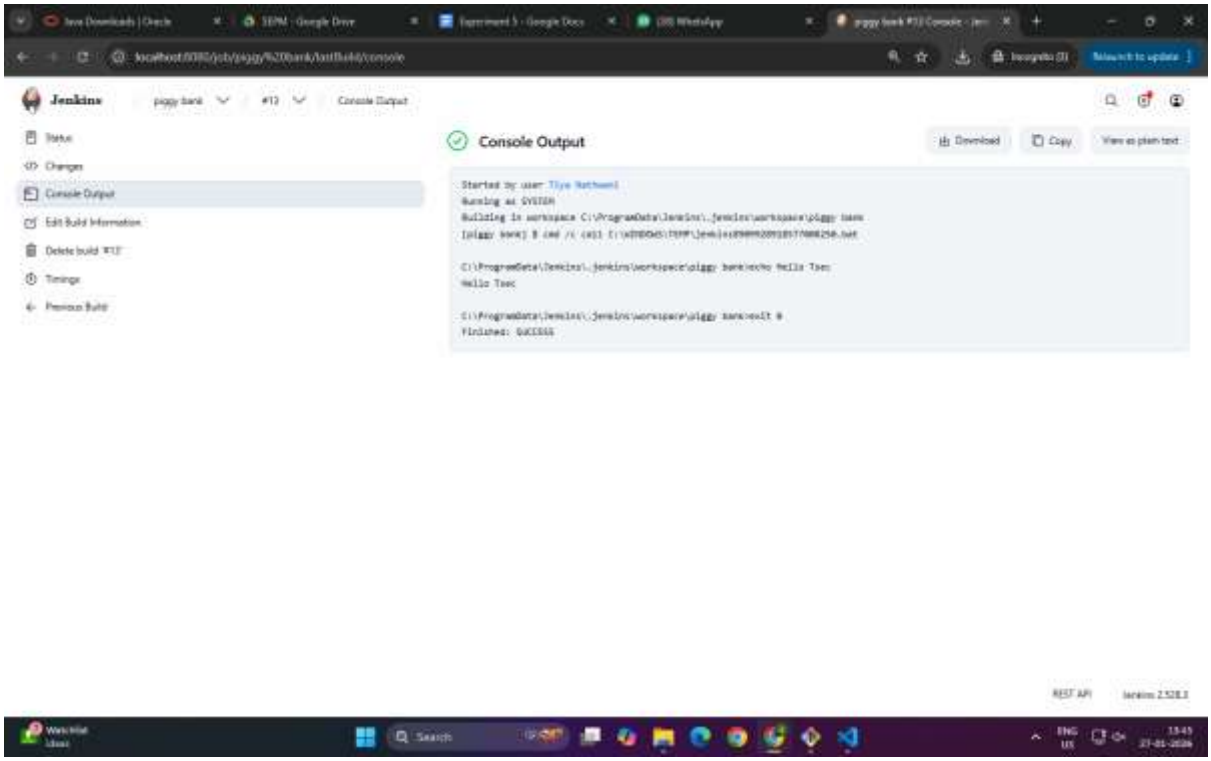

Jenkins
New Item
Pipeline
Freestyle project
Multi-configuration project
Folder
Multibranch Pipeline
OK


Jenkins
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Save
Apply

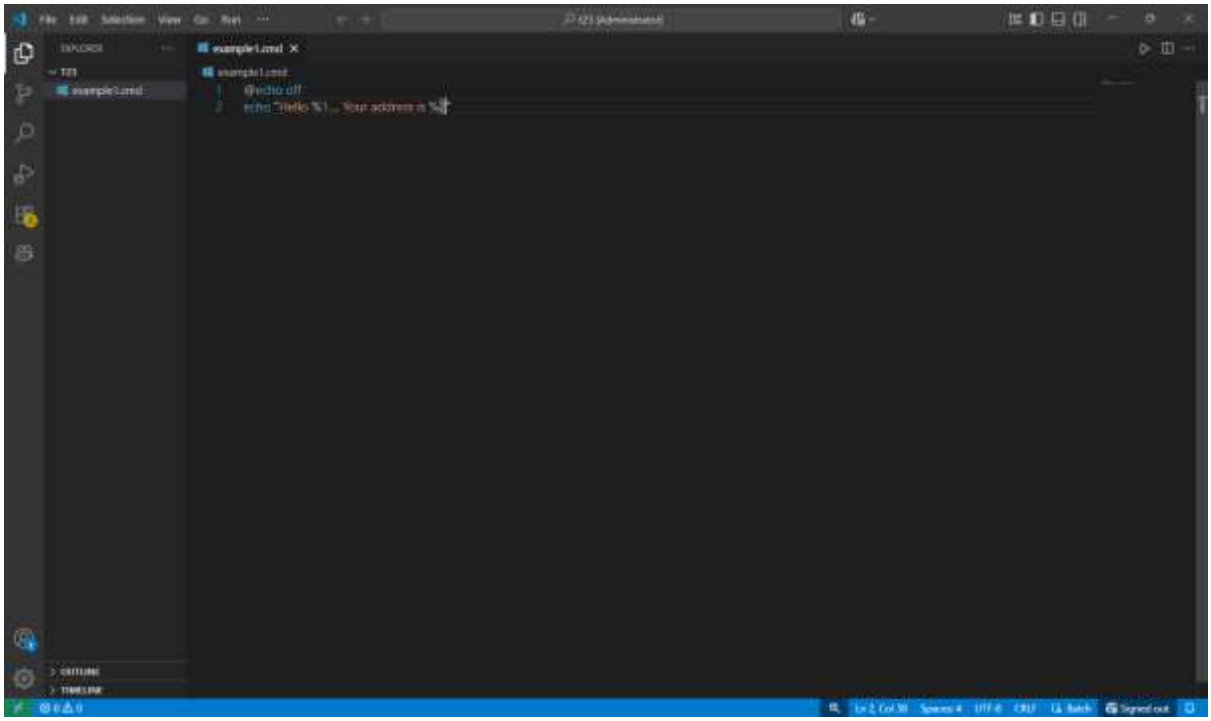

Jenkins
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Command
Advanced
Add build step
Post-build Actions
Add post-build action
Save
Apply
Saved

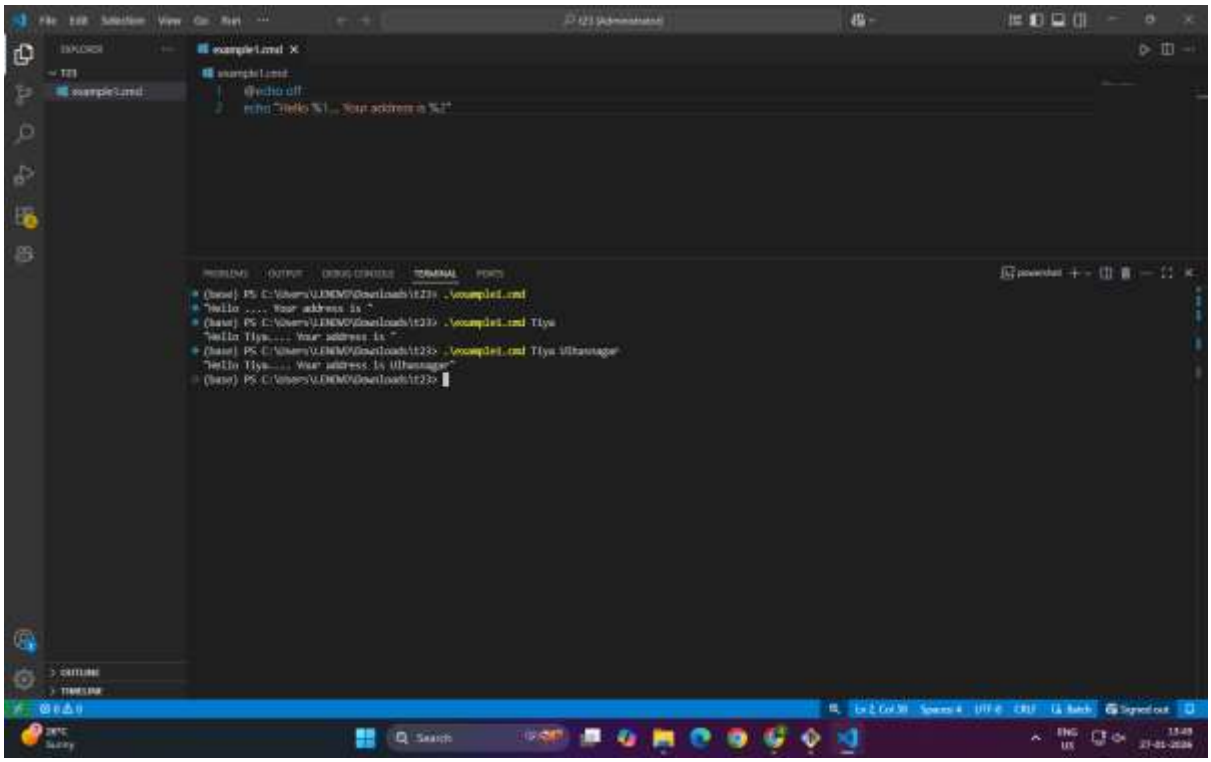

Jenkins
piggy bank
Configuration
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Delete workspace before build starts
Add timestamps to the Console Output
Build Steps
Execute Windows batch command
Command
Advanced
Save
Apply

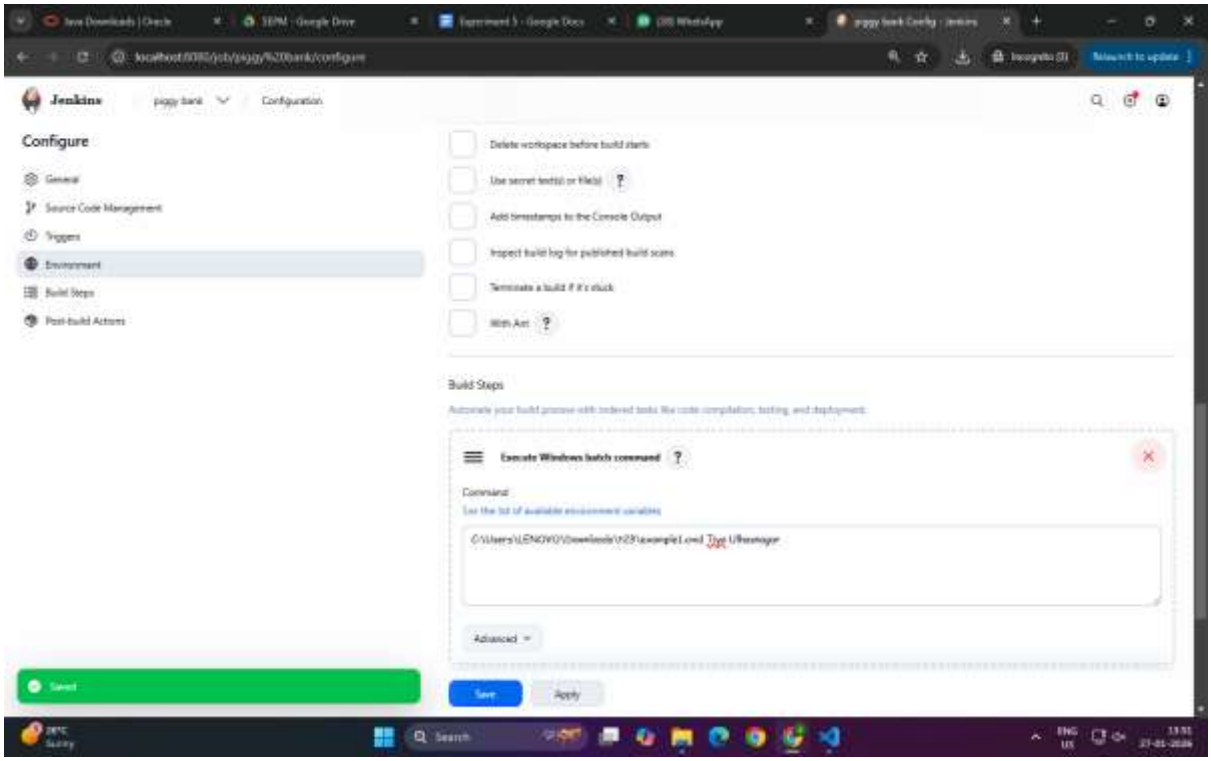

Jenkins
piggy bank
Status
Changes
Workspace
Build Now
Configure
Delete Project
Rename
piggy bank
Permalinks
Builds

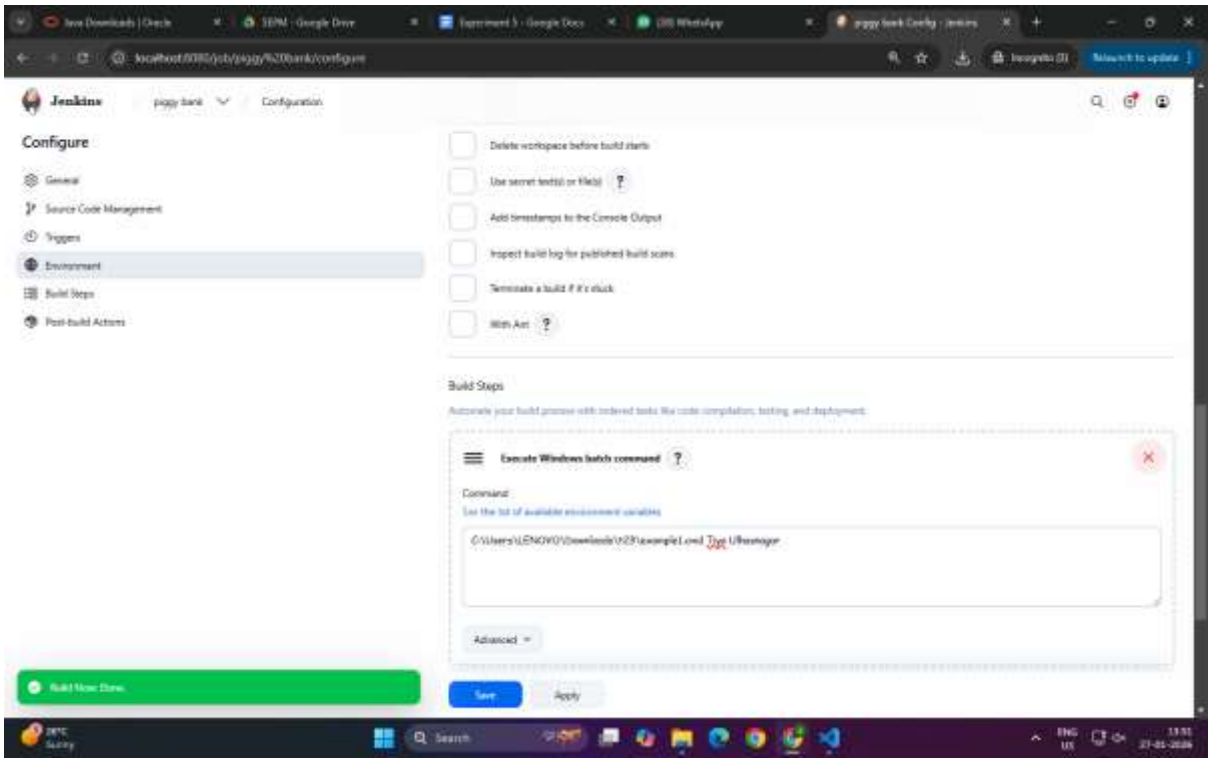

Jenkins
Console Output
Status
Changes
Console Output
Edit Build Information
Delete build
Timings
Previous Build
Download
Copy
Started by user
Running as SYSTEM
Hello Tsec
Finished: SUCCESS
REST API

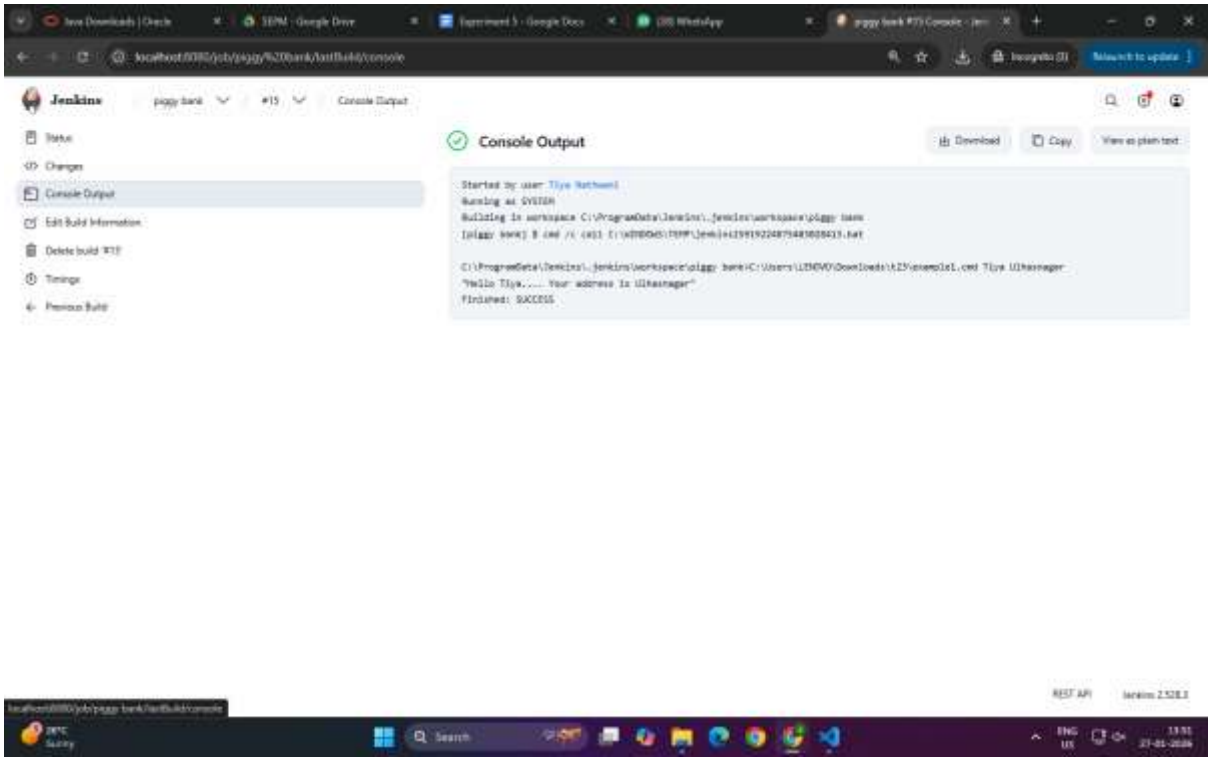
1.2



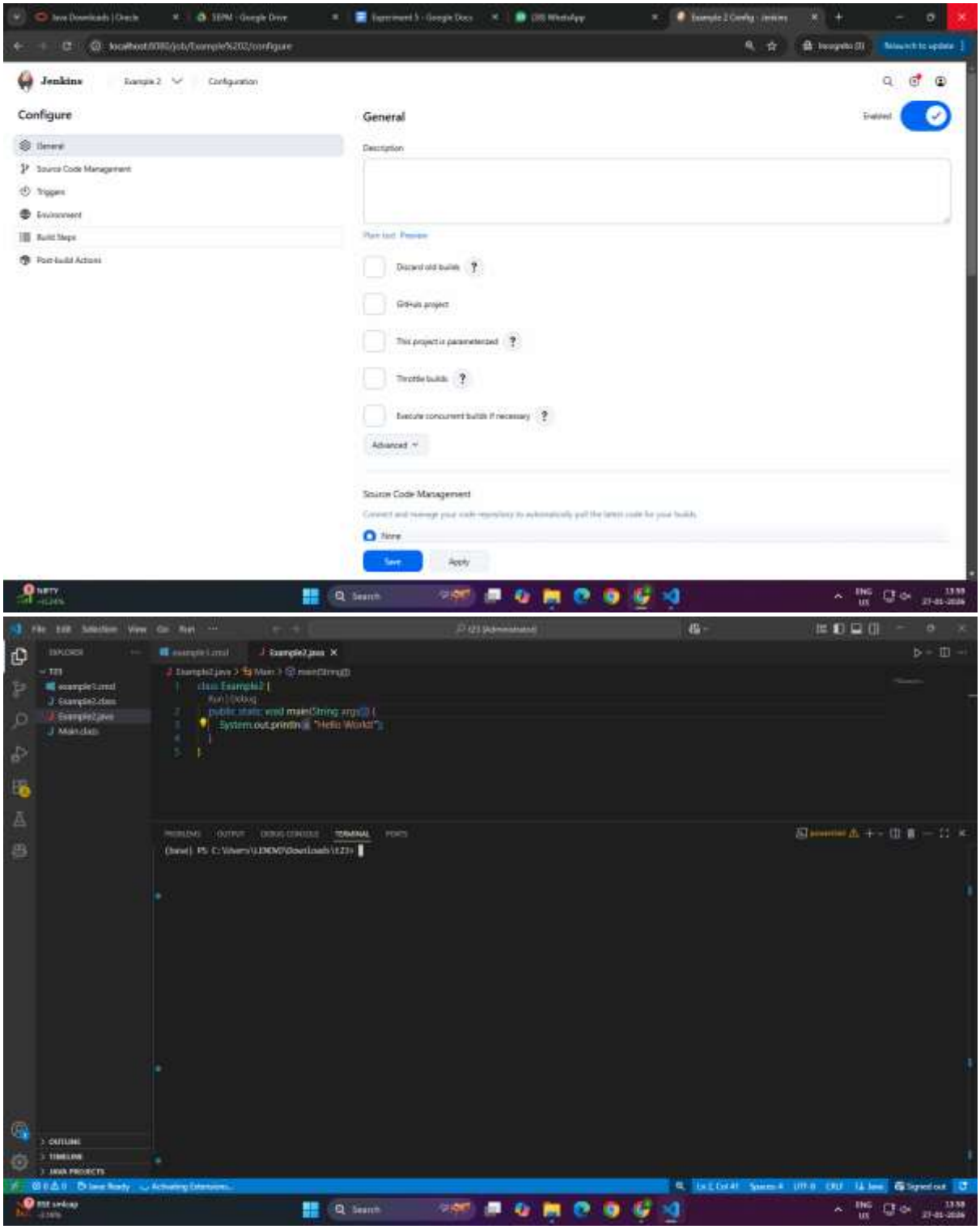


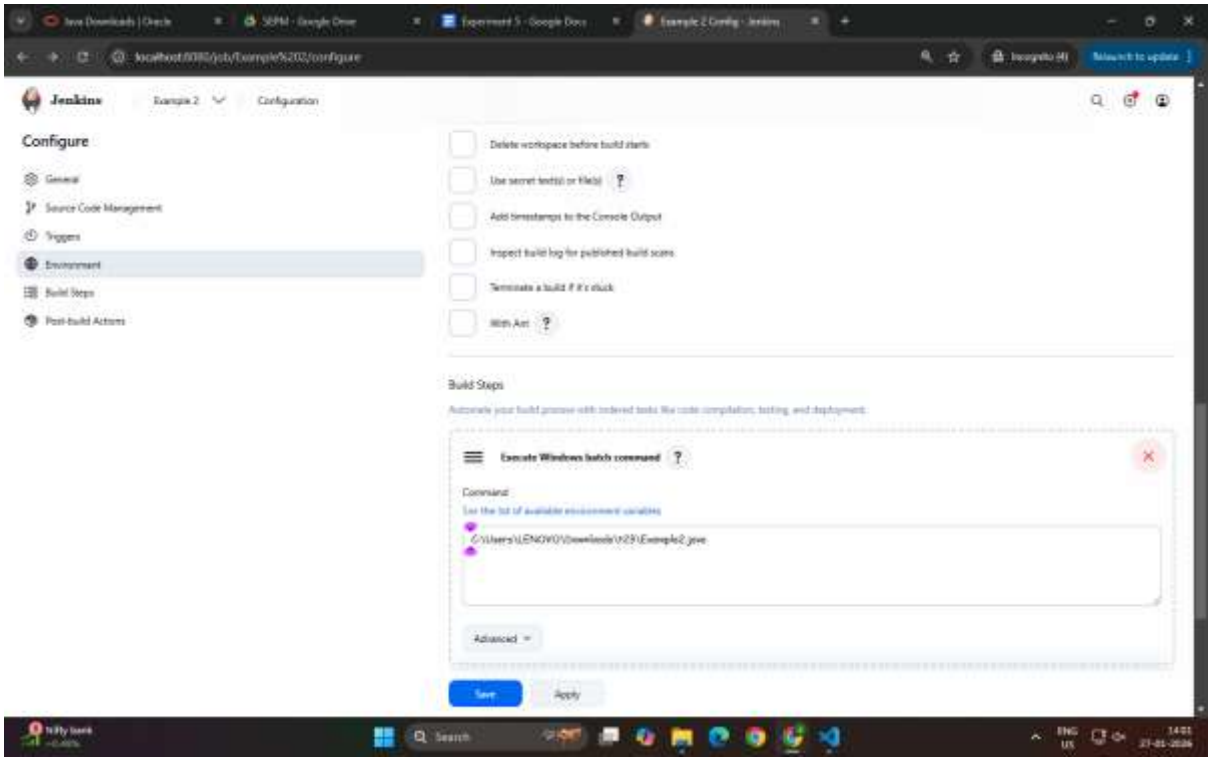

Jenkins
Configuration
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Build Steps
Execute Windows batch command
Command
Advanced
Save
Apply

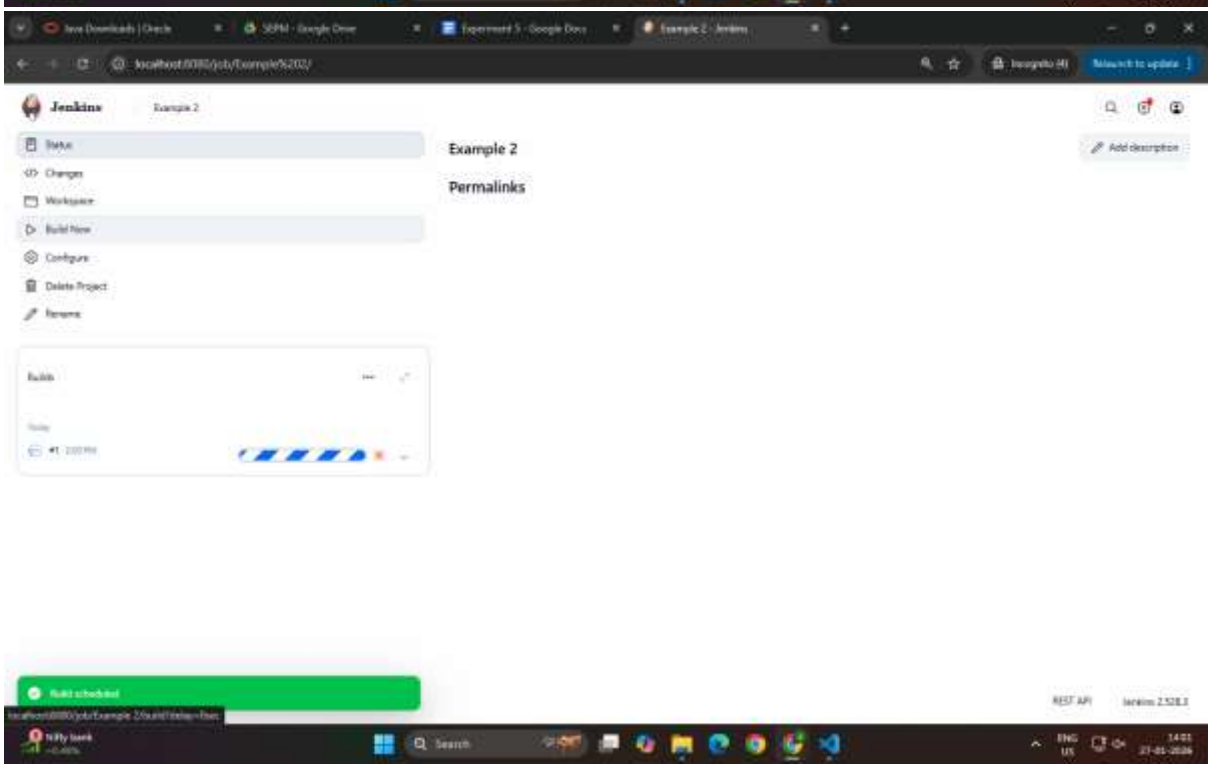

Jenkins
Configuration
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Build Steps
Execute Windows batch command
Command
Advanced
Save
Apply

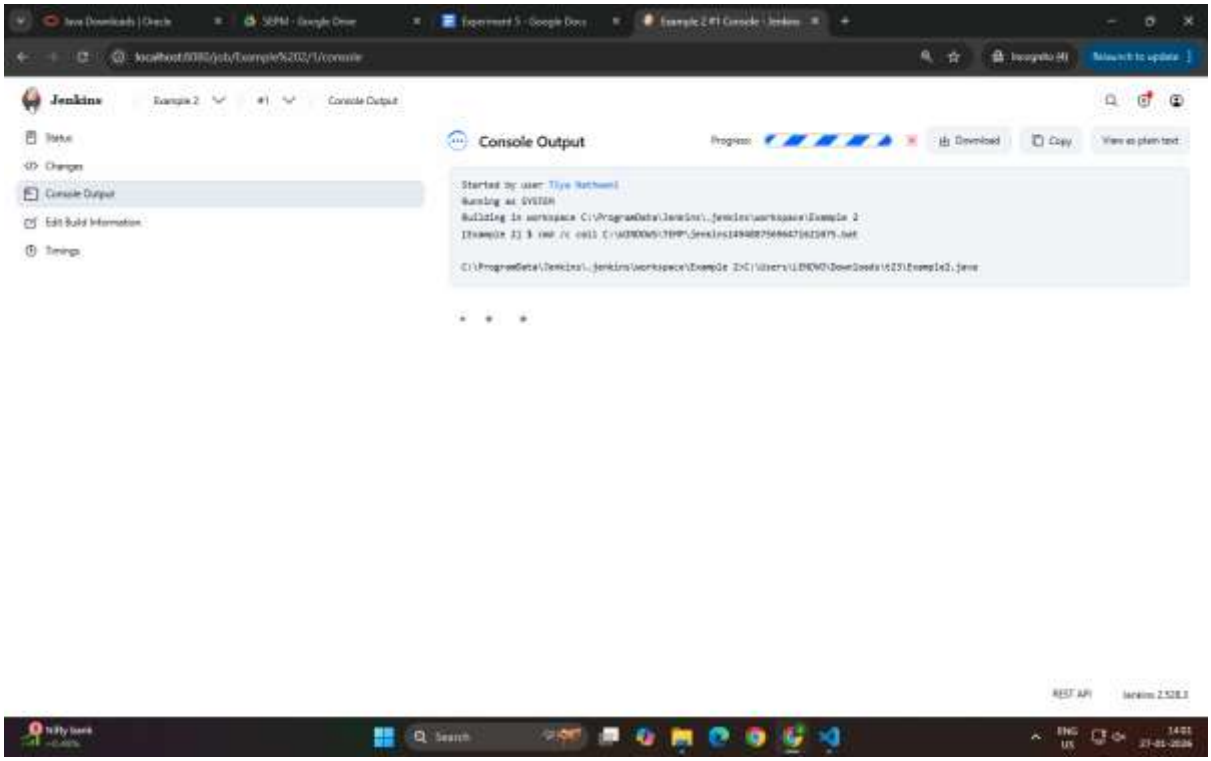

Jenkins
Console Output
Download
Copy
Status
Changes
Console Output
Edit Build Information
Delete build '#15'
Timings
Previous Build
Started by user Tiya Nathwani
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\.jenkins\workspace\piggy bank
"Hello Tiya..... Your address is Ulhasnagar"
Finished: SUCCESS
REST API

Example 2

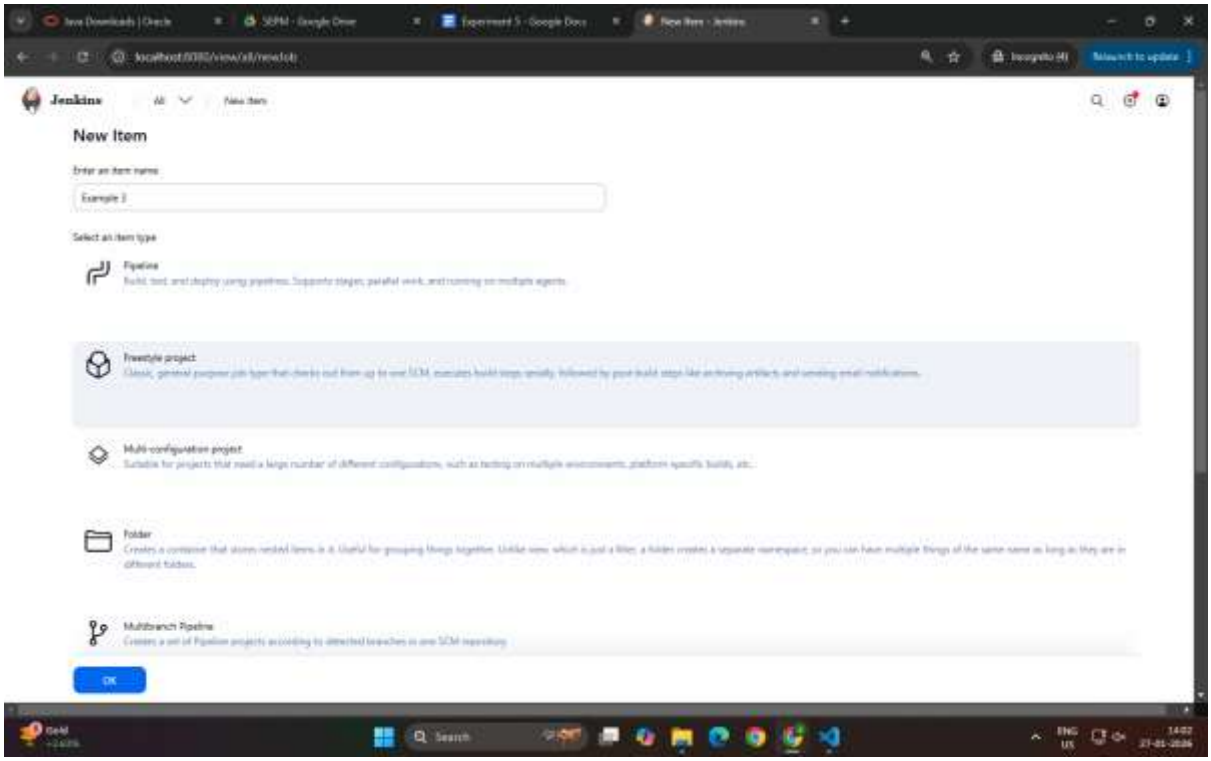

Jenkins
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Description
Discard old builds
GitHub project
This project is parameterized
Throttle builds
Execute concurrent builds if necessary
Advanced
Source Code Management
None
Save
Apply
class Example2 {
public static void main(String args[]) {
System.out.println("Hello World");
}
}

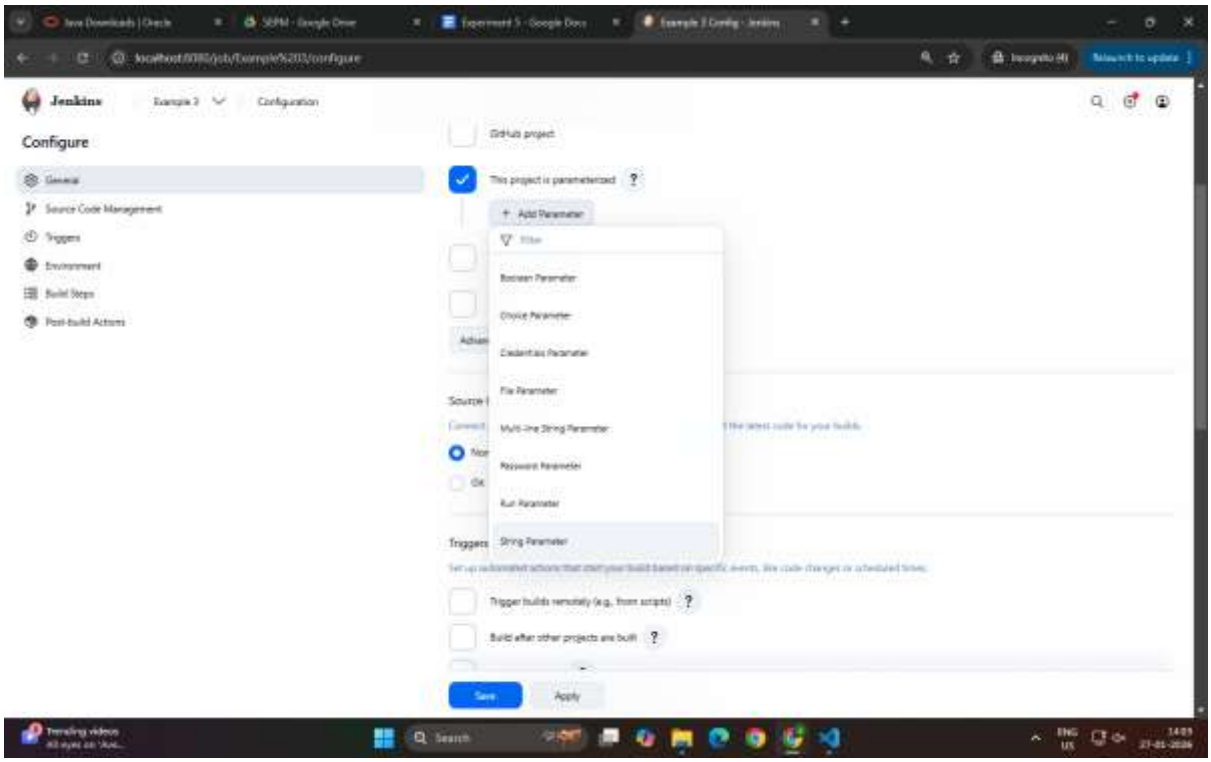

Jenkins
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Build Steps
Execute Windows batch command
Command
Advanced
Save
Apply

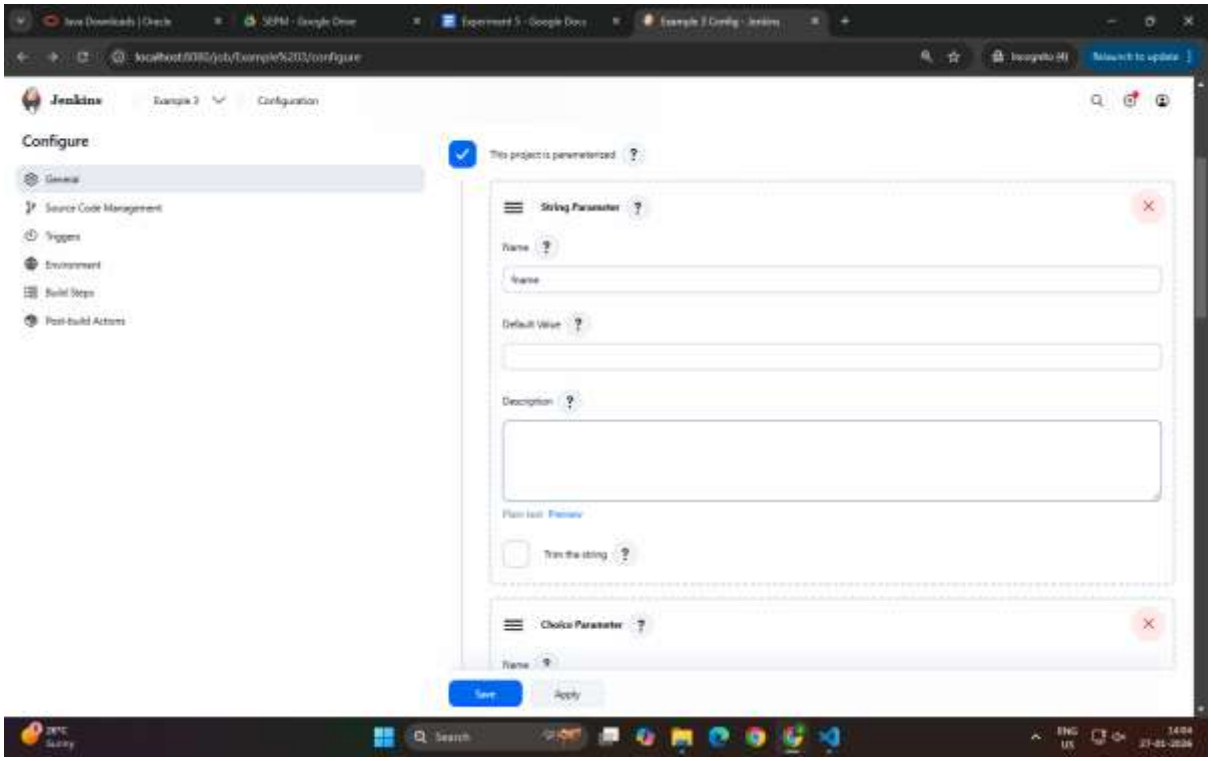

Jenkins
Example 2
Changes
Workspace
Build Now
Configure
Delete Project
Rename
Example 2
Permalinks
Add description
Build scheduled

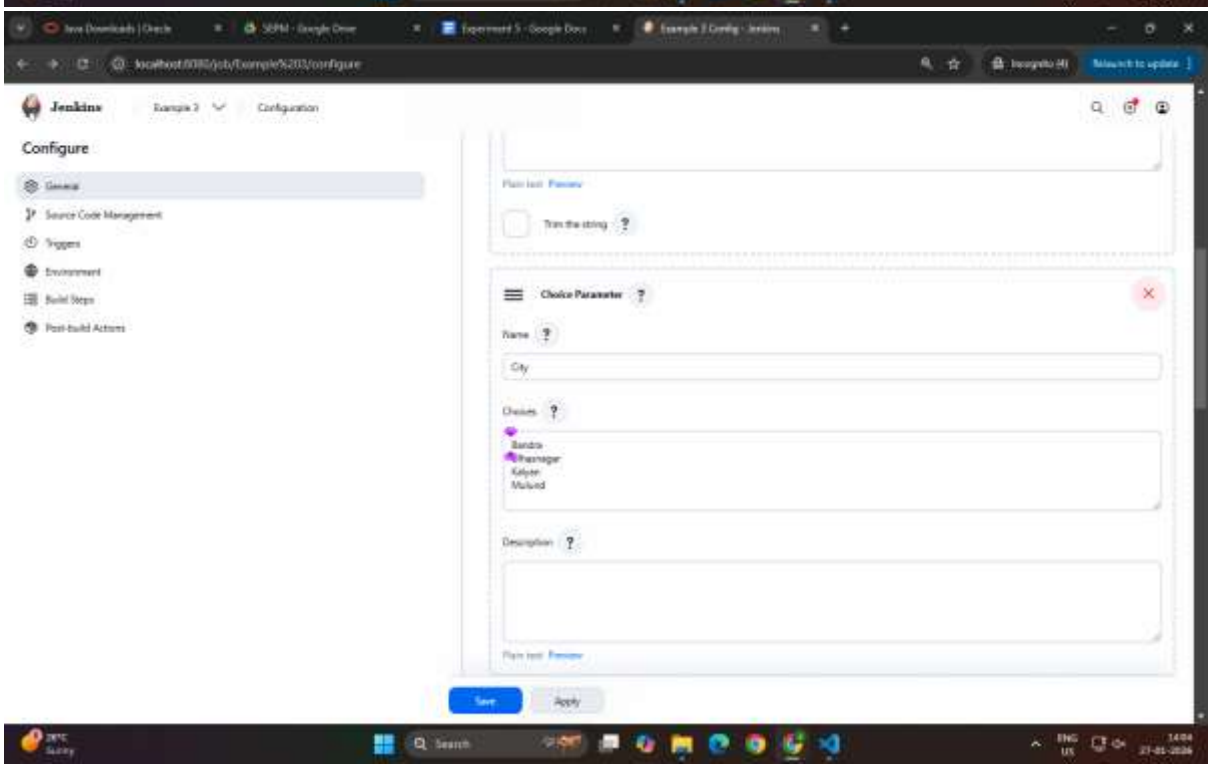

Jenkins
Console Output
Download
Copy

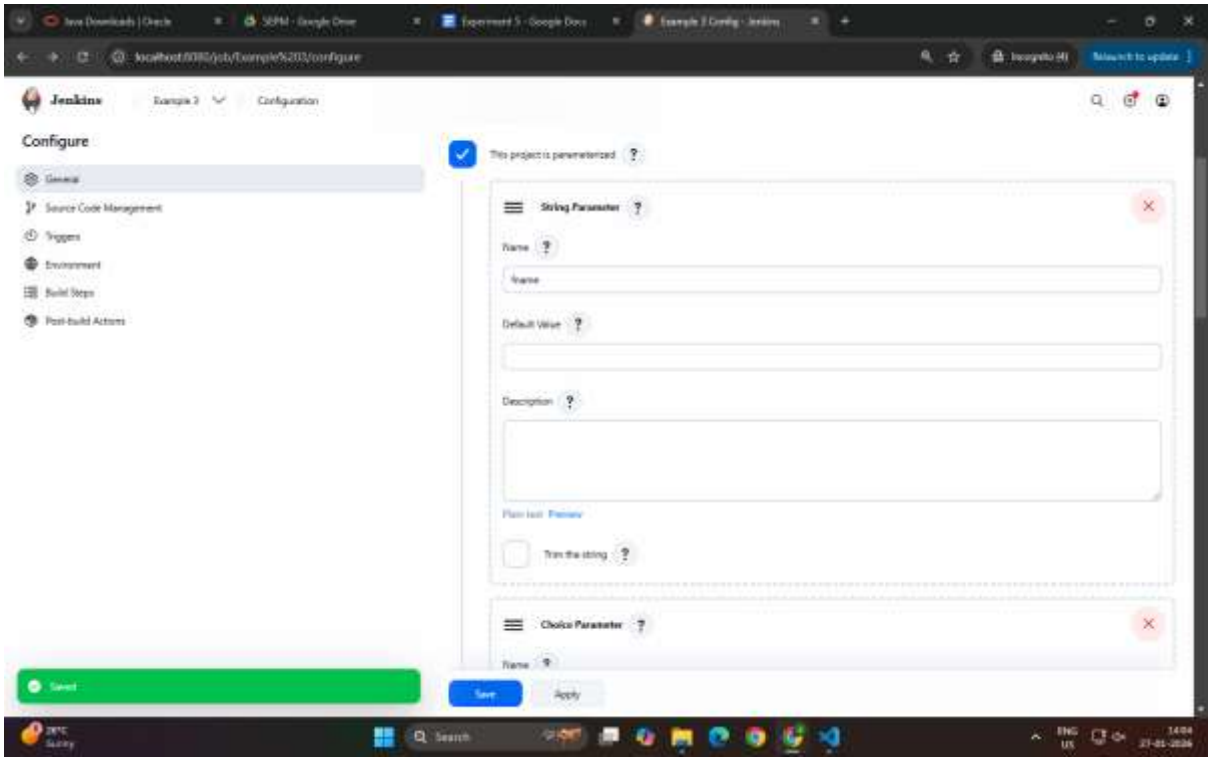
Example 3

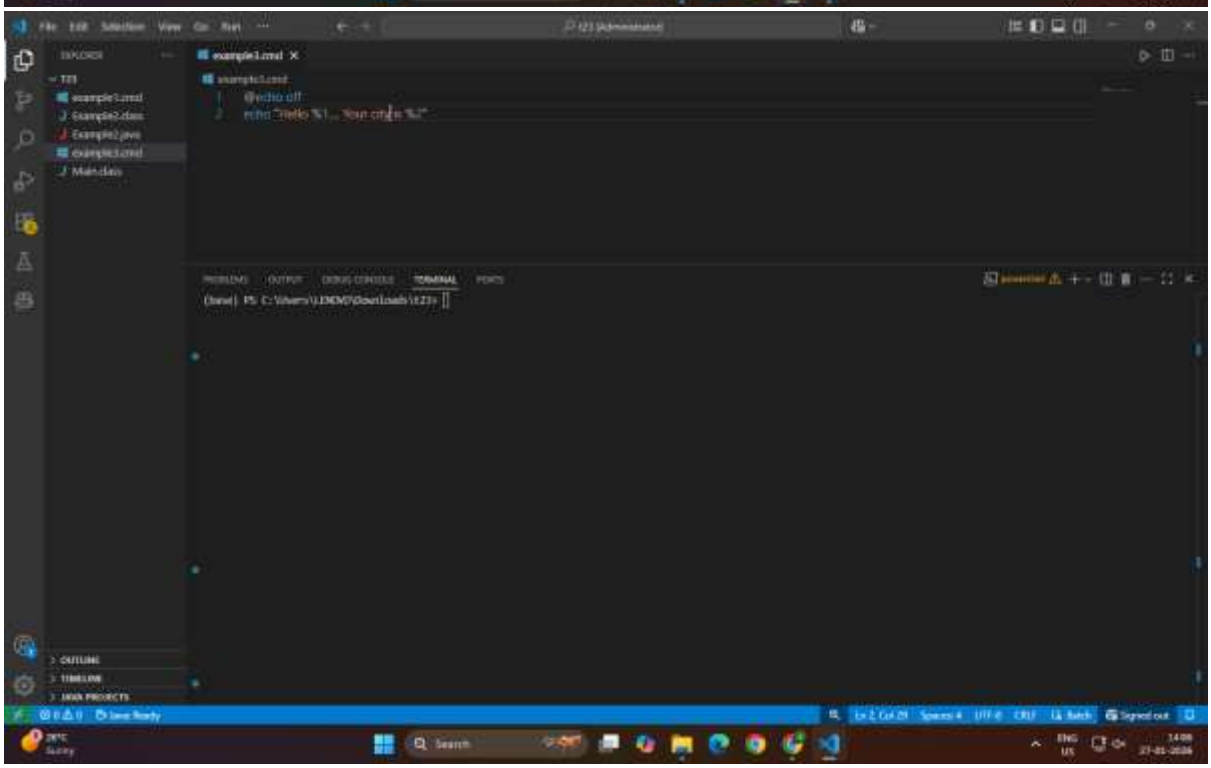

Jenkins
New Item
Enter an item name
Example 3
Select an item type
Pipeline
Freestyle project
Multi-configuration project
Folder
Multibranch Pipeline
OK

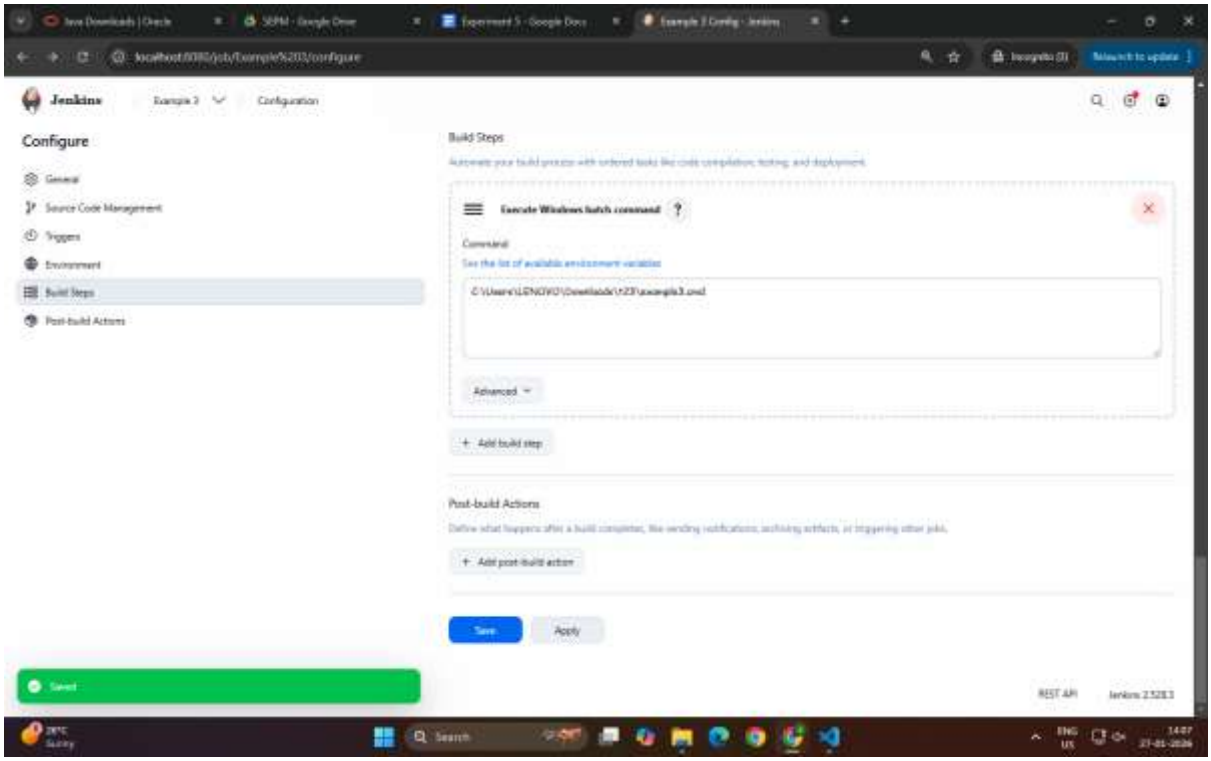

Jenkins
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
This project is parameterized
Add Parameter
Boolean Parameter
Choice Parameter
File Parameter
Password Parameter
Run Parameter
String Parameter
Trigger builds remotely (e.g., from scripts)
Build after other projects are built
Save
Apply

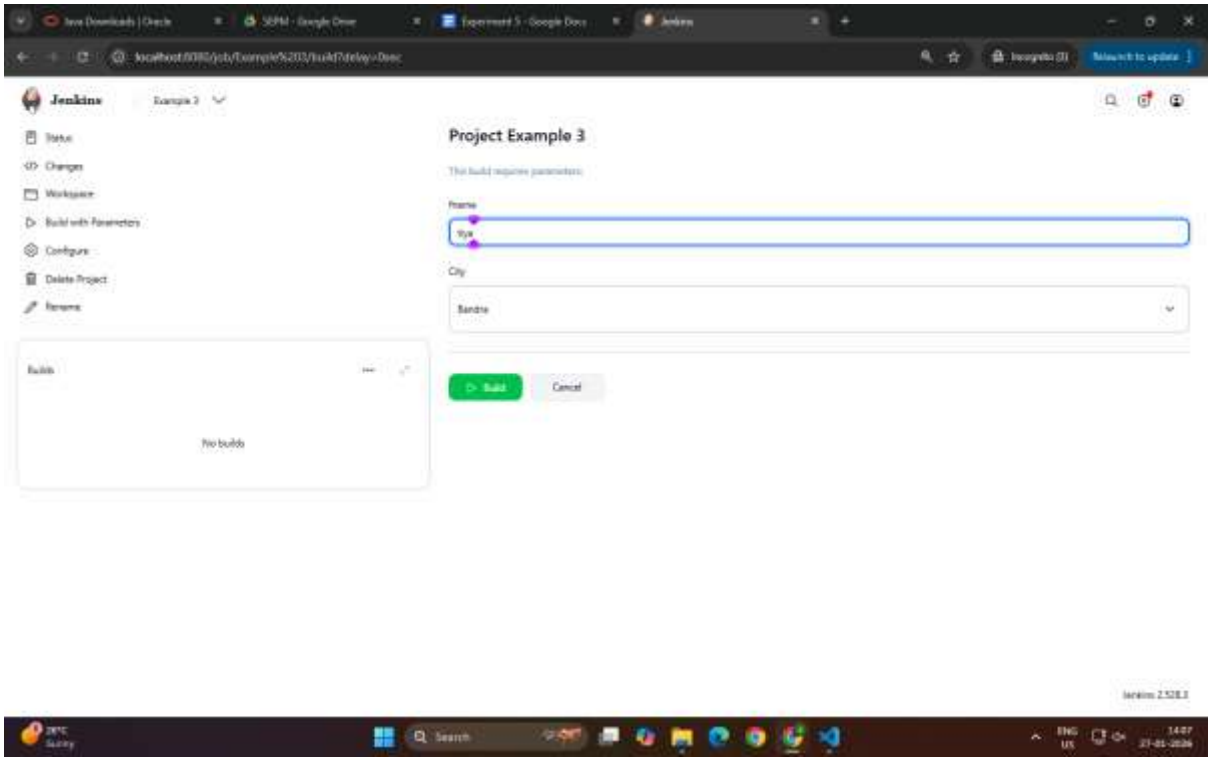

Jenkins
Configure
String Parameter
Choice Parameter
Save
Apply

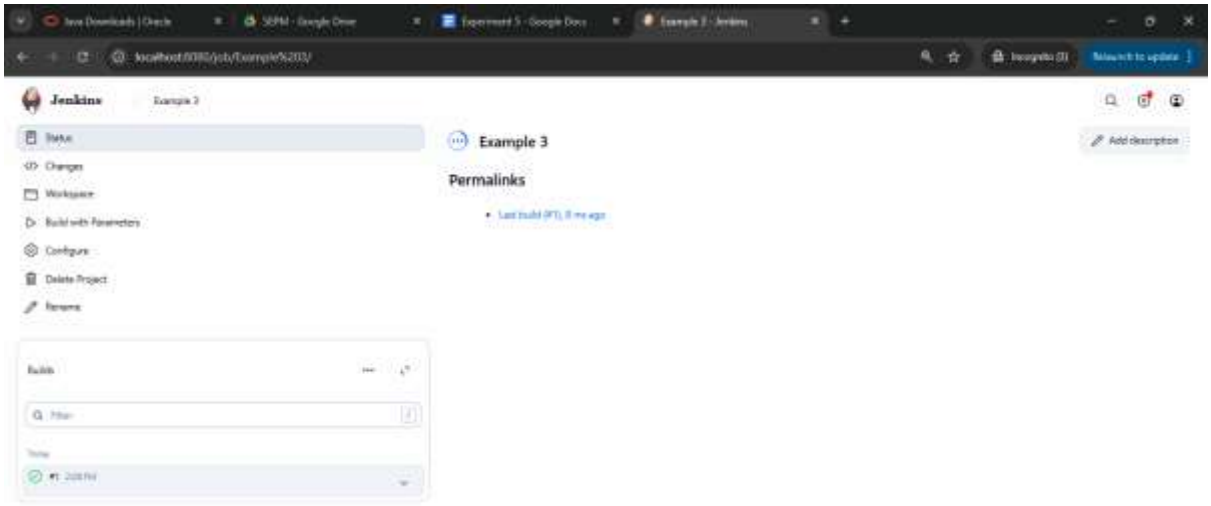

Jenkins
Configure
Choice Parameter
City
Save
Apply

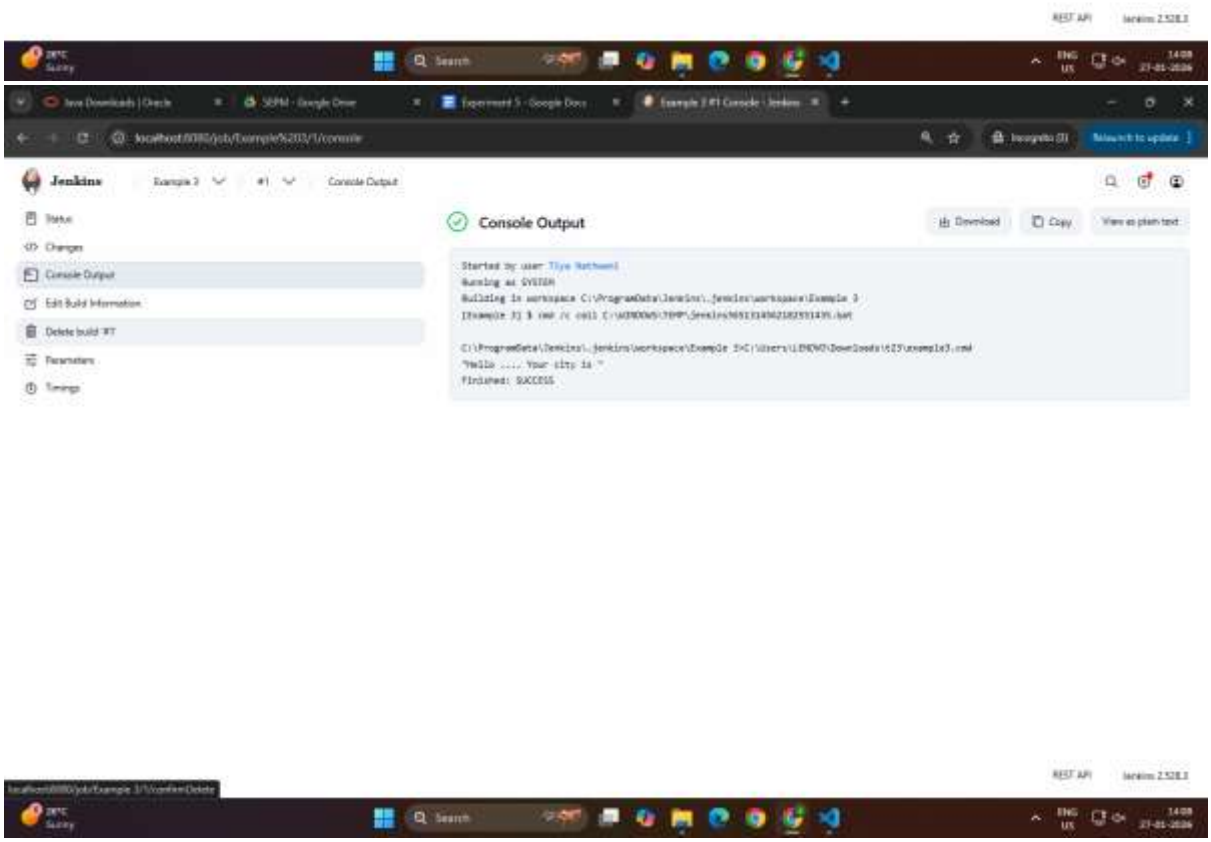

Jenkins
Configure
This project is parameterized
String Parameter
Choice Parameter
Save
Apply


example1.cmd
@echo off
TERMINAL

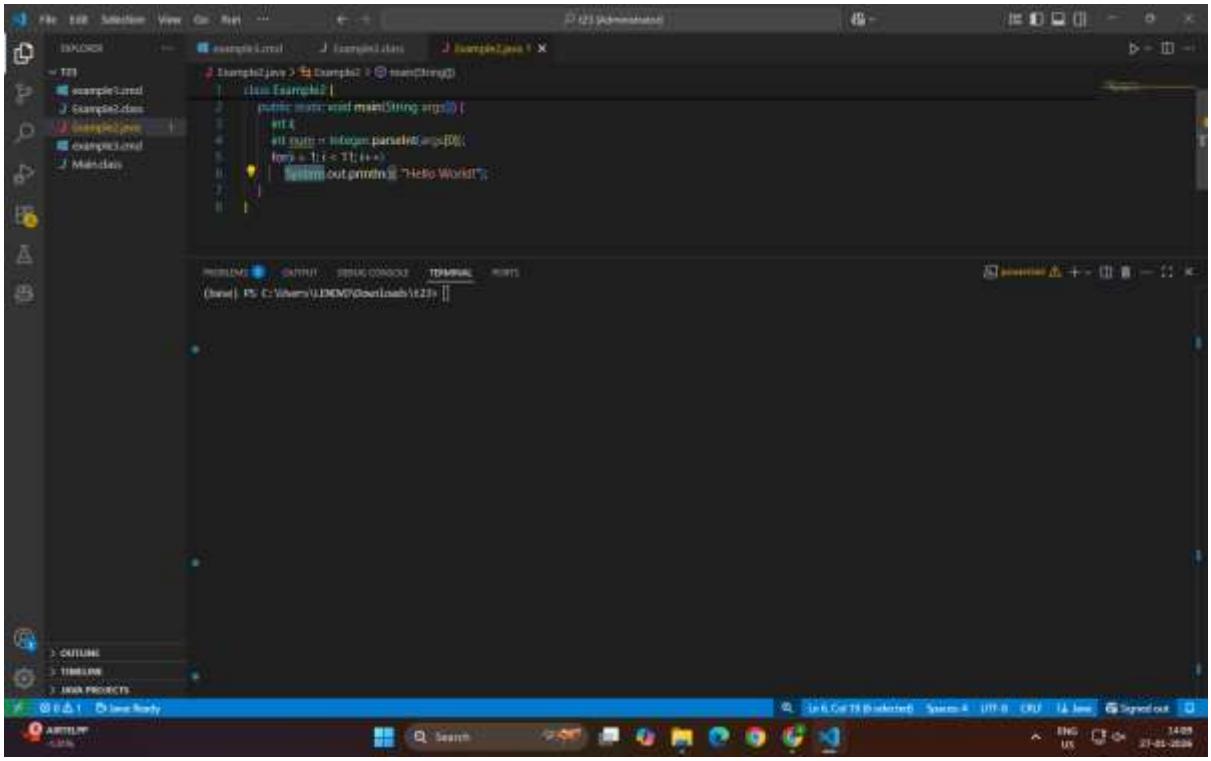

Jenkins
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Build Steps
Execute Windows batch command
Command
Advanced
Add build step
Post-build Actions
Add post-build action
Save
Apply
Saved
REST API

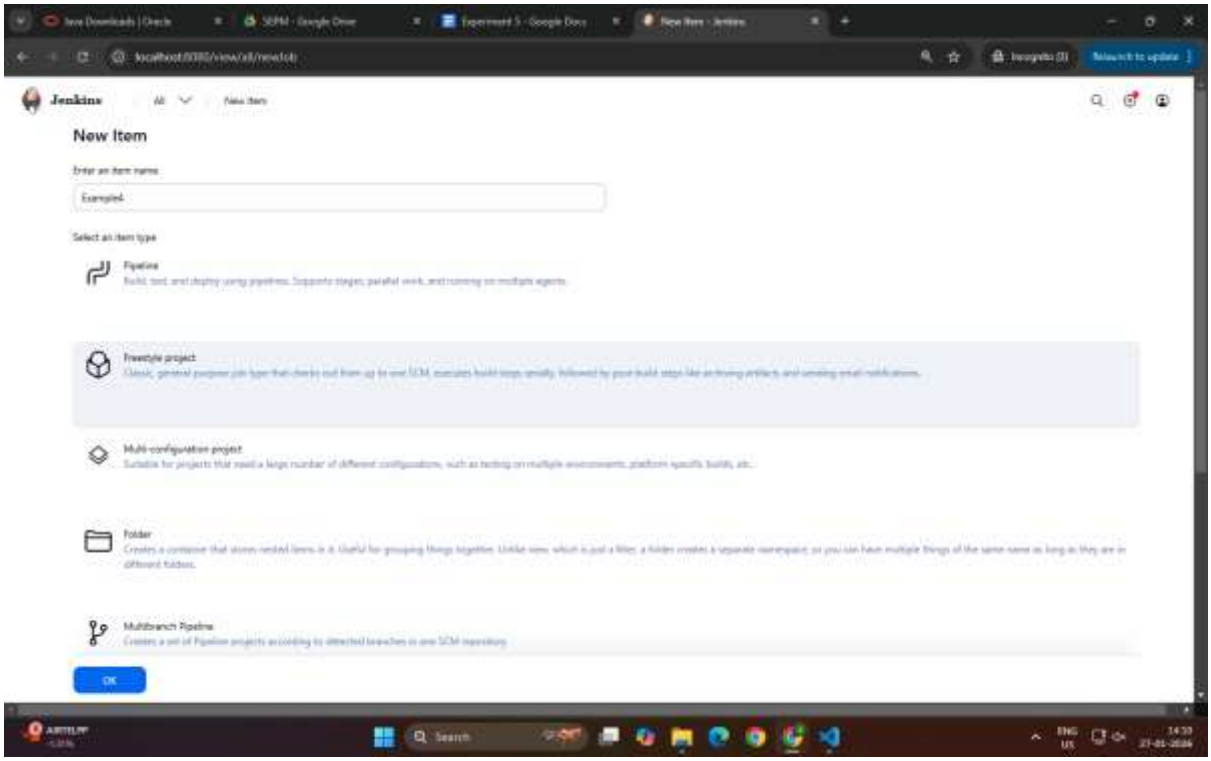

Jenkins
Status
Changes
Workspace
Build with Parameters
Configure
Delete Project
Rename
Project Example 3
Name
City
Build
Cancel
No builds

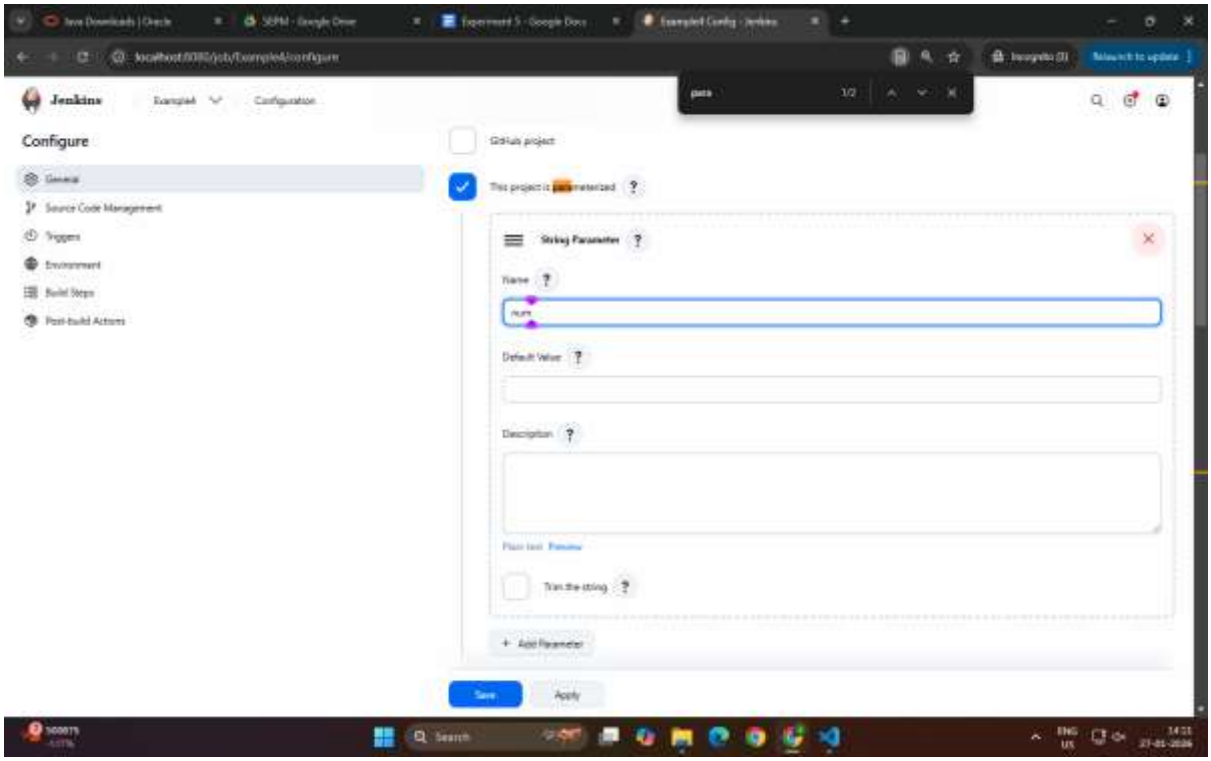


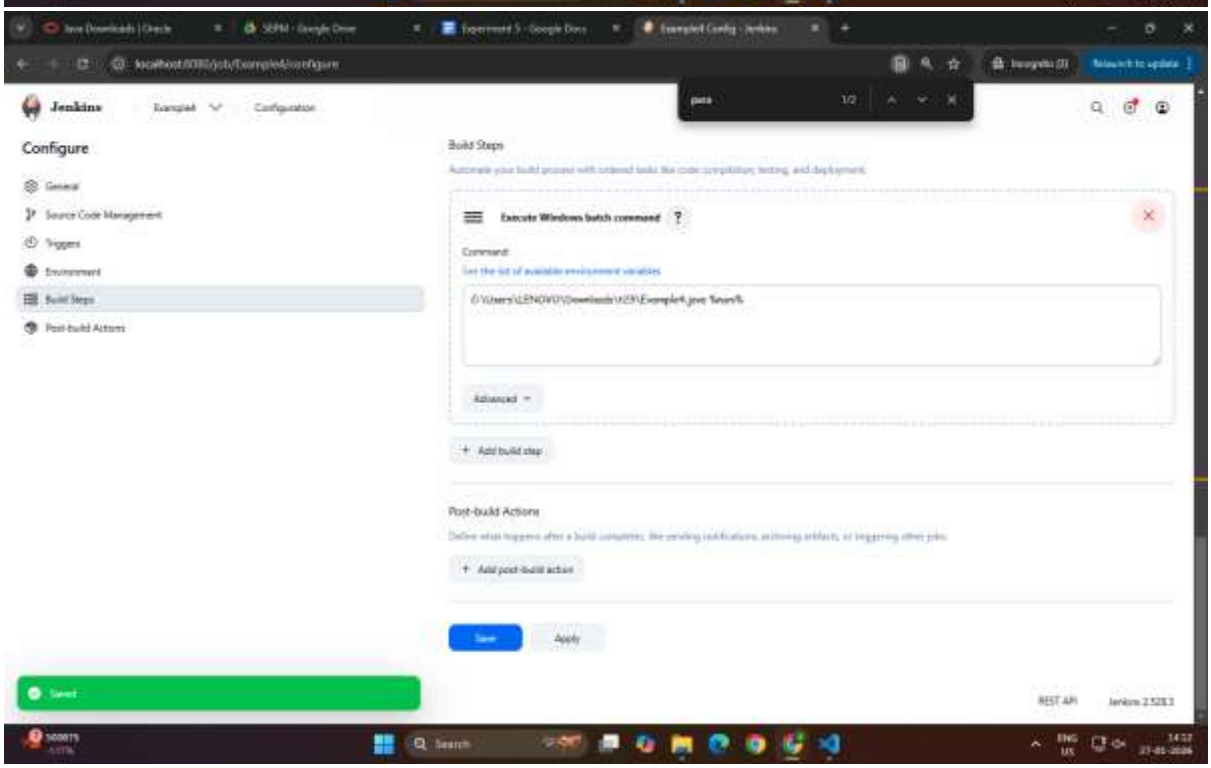


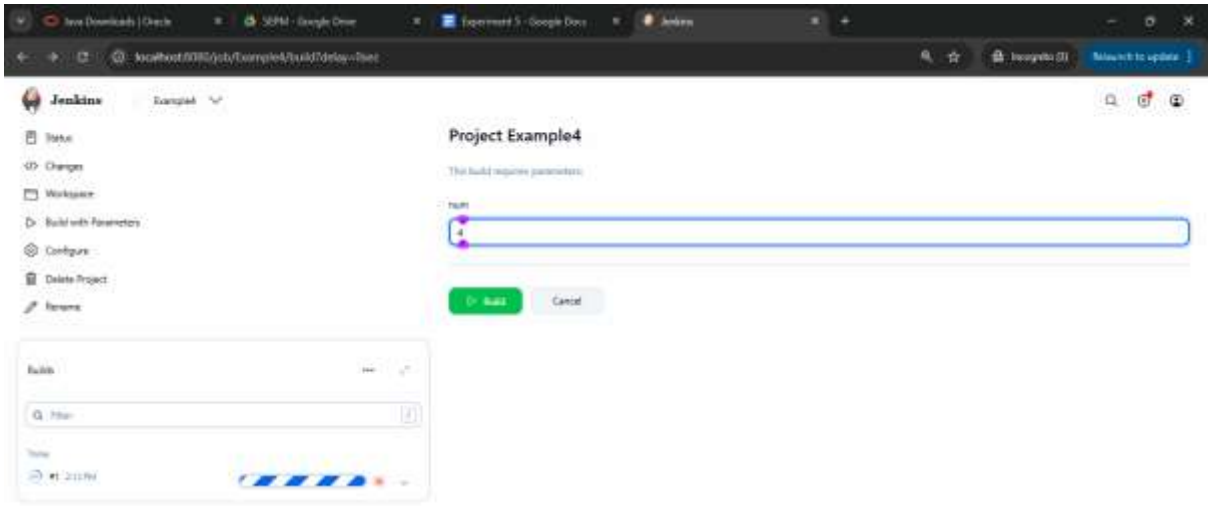
3.2

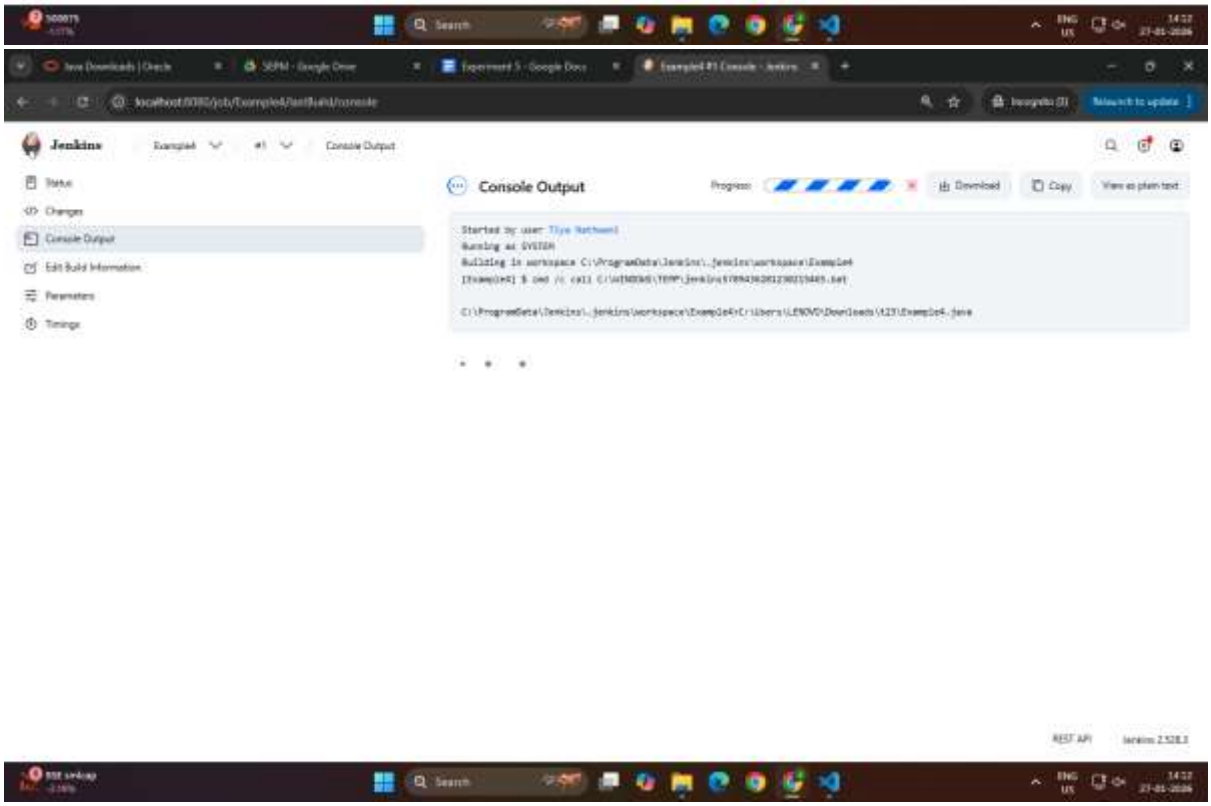



Jenkins
New Item
Enter an item name
Example1
Select an item type
Pipeline
Freestyle project
Multi-configuration project
Folder
Multibranch Pipeline
OK


Jenkins
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
String Parameter
Name
num
Default Value
Description
Trim the string
Add Parameter
Save
Apply


Jenkins
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Build Steps
Execute Windows batch command
Command
Advanced
Add build step
Post-build Actions
Add post-build action
Save
Apply
Saved
REST API


Jenkins
Project Example4
Build
Cancel


Jenkins
Console Output
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